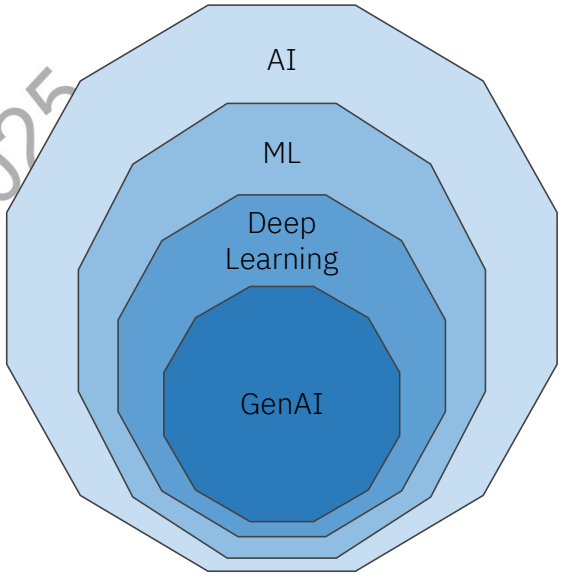
Introduction to AI

Week 7: Perceptron learning

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Data & AI Leader, IBM ASEAN

Tran Dang An
Assistant

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Objectives

- Explain the concept of artificial neuron networks (ANN)
- Understand the structure and operation of a single-layer perceptron
- Describe how Multilayer Perceptrons (MLP) work when adding more hidden layers enables models to learn more complex patterns.

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AI Hierarchy

Artificial Intelligence

W1: Basic concepts, history of AI
W2: Knowledge representation and logic
W3: Searching algorithms (pathfinding and text)

Machine Learning

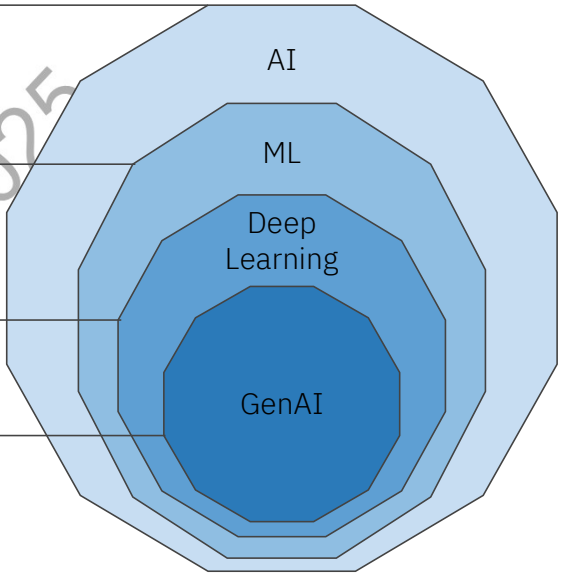
W4: Statistical learning
W5: Supervised machine learning
W6: Unsupervised machine learning

Deep Learning

W7: Perceptron learning and neural network learning
W8: Deep learning

Generative AI

W9 + W10: Modern AI and development in real world applications (LLM & GenAI)



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Contents

- From ML to ANN
  - Single Perceptron Model
  - Multi-Layer Perceptrons (MLP)

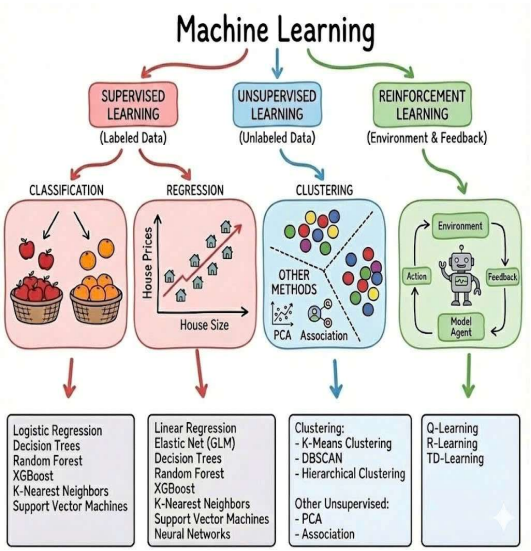
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Machine Learning Recap

- Supervised Learning:
  - **Goal**: Predict future outcomes based on past examples.
  - **Data**: Labeled
  - **What we did**: Regression (Linear Regression), Classification (KNN).
- Unsupervised Learning:
  - **Goal**: Find hidden patterns or structures.
  - **Data**: Unlabeled (Raw data only)
  - **What we did**: Clustering (K-Means), Dimensional Reduction (PCA).



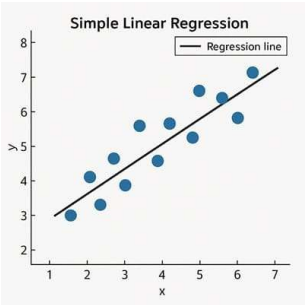
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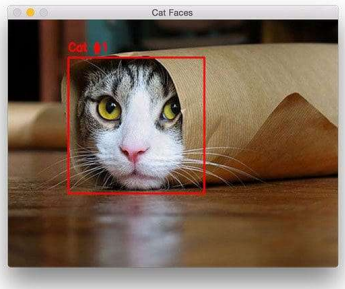
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The Limits of Machine Learning

- We used algorithms like Linear Regression and K-Means, which are excellent at finding clear, simple relationships:
  - Example: like drawing a "best fit" straight line through data points.
- The problem: real-world data is rarely straight.
  - Example: Identifying a cat in a photo isn't about finding a straight line; it's about shapes, textures, and context.





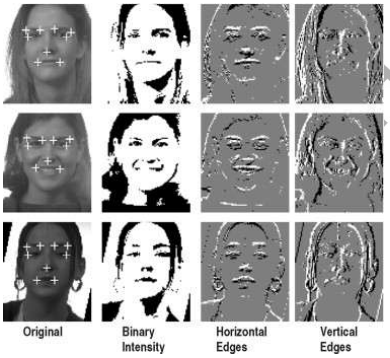
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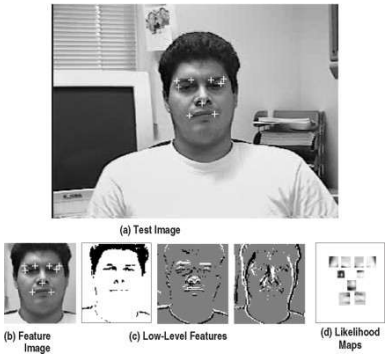
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Example of Feature Engineering for Face Recognition

- For complex tasks (recognizing a face), impossible to manually list every features
  - Example: "distance between eyes," "curvature of chin“
- Need a better way (e.g. good AI model) which can learn the feature engineering better than traditional way



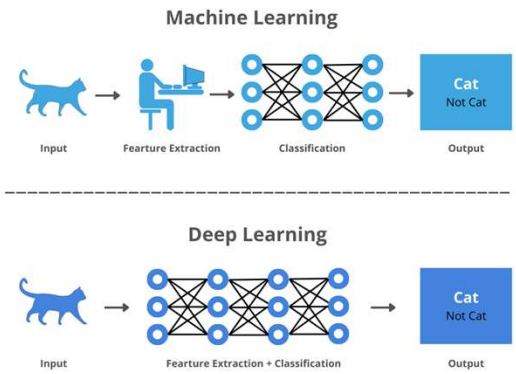


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Enhancement of "Feature Engineering“ in DL

- In Machine Learning
  - Humans must use domain knowledge to extract specific features
  - The algorithm acts only on those prepared features. If the human misses a feature, the model cannot learn it.
- In Deep Learning
  - The initial layers of the network automatically learn to identify simple features (edges, curves), and deeper layers combine them into complex features (eyes, faces).
  - The model figures out what to look for without human intervention.

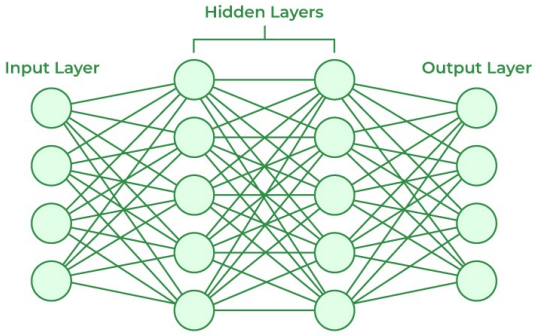


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Artificial Neural Network (ANN)

- A computing system made up of simple, interconnected processing units (neurons) that together solve complex problems.
- Key Concept:
  - A single neuron is mathematically very simple (just simple addition and multiplication).
  - But when you stack layers of them together, they can solve incredibly difficult problems.

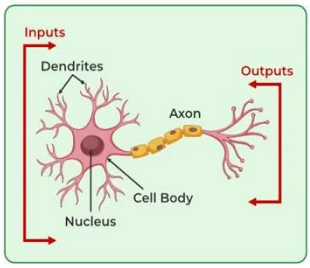


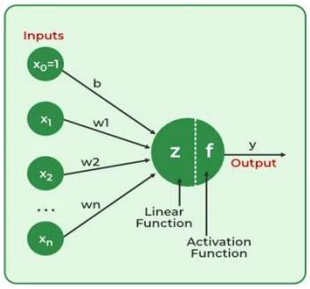
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The Biological vs. Artificial Neuron





Biological Neuron

- Dendrites (Receivers): Collect signals from other neurons.
- Soma (Processor): Sums the signals to decide if the neuron should fire.
- Axon (Transmitter): Sends the electrical signal to the next neuron.
- About 80-100bil neurons in our brain

Artificial Neuron

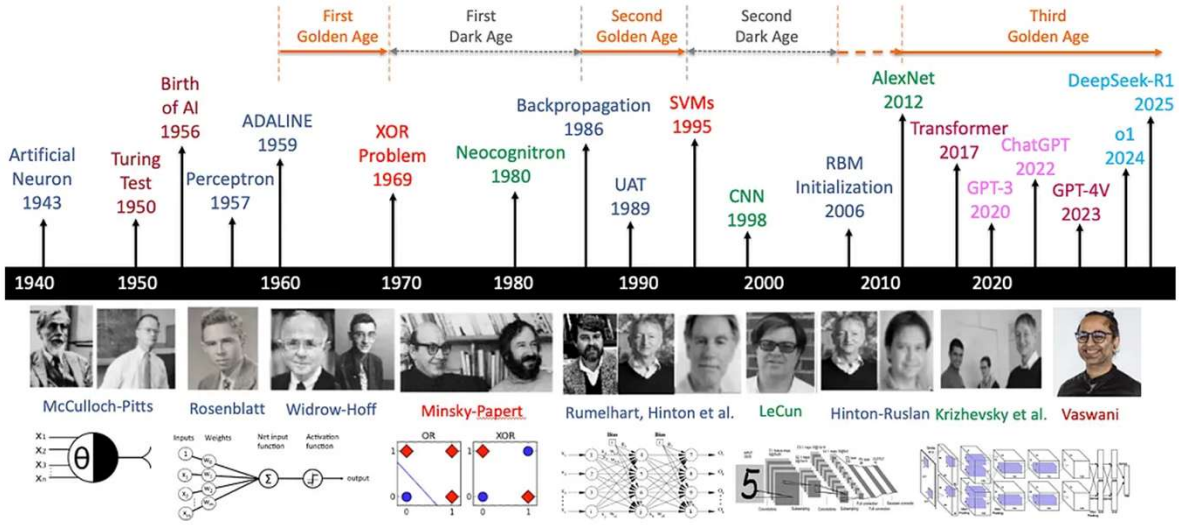
- Input Layer: Receives data.
- Weights: Assigns importance to data.
- Summation: Aggregates the signals.
- Activation Function: Decides the output.
- Output: Linear or logistic
- Small perceptron models: 100 neurons, multi-layered can be up to 1-10k neurons
- Complicated DL models like CNN and LLM can be few mil neurons

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Timeline: from ANN to DL and now Agentic AI



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Contents

- From ML to ANN
- Single Perceptron Model
- Multi-Layer Perceptrons (MLP)

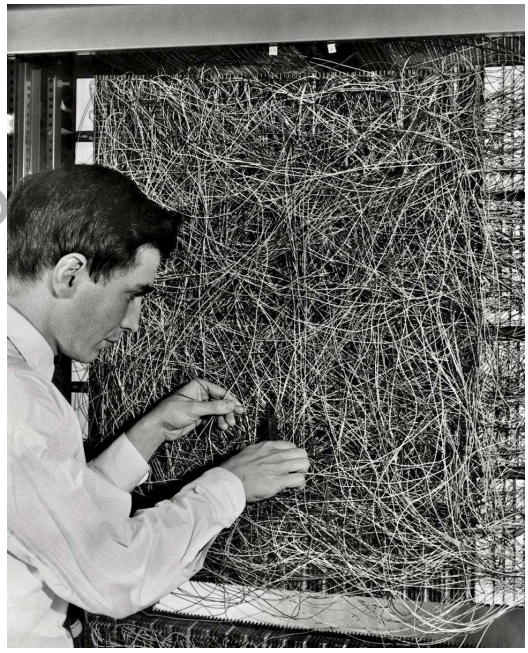
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A Perceptron

- A Perceptron is the simplest form of a neural network, the fundamental building block of deep learning models.
- Invented by Frank Rosenblatt in 1957 at the Cornell Aeronautical Laboratory.
- Image: *Frank Rosenblatt with a Mark I Perceptron computer in 1960*



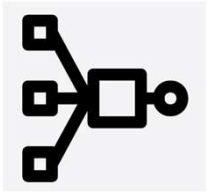
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Perceptron Concept & History

- Perceptron is an atomic of neural network.
- Originally, the output of perceptron is 0 or 1 (later it is improved to S-curve using real number)
- Early use-cases are to solve the logic functions like AND, OR
- 1960: High expectations to solve many problems including image, language. However, they found that single perceptron can solve the linear problem only (XOR problem in 1969) -> AI winter
- 1980: the MLP was introduced with Backpropagation enabled training networks with hidden layers
- 2010s+ with DL and later Transformer all leverage the basic concept of perceptron



Sample icon designed by M. Oki Orlando

The **Perceptron** comes from
- "Perceive" recognize patterns
- "-tron" a machine or electronic device

Perceptron → MLP → Deep Learning → Transformers

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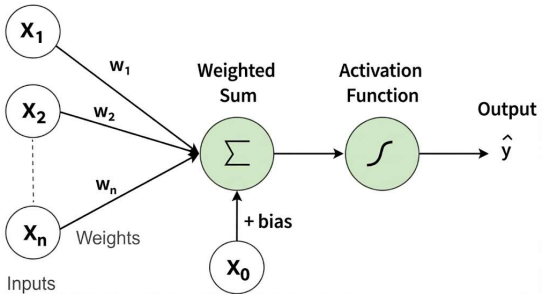
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The Anatomy of a Perceptron

- A Perceptron is composed of:

1. **Input Layer**: Receives data.
2. **Weights**: Assigns importance to data.
3. **Summation**: Aggregates the signals.
4. **Activation**: Decides the output.
5. **Output:** Produces linear (real number) or logistic (0|1)



- Data flows in one direction:
  - Input →Processing →Output.
  - This is why it's called a Feed-Forward neural network (FFNN)

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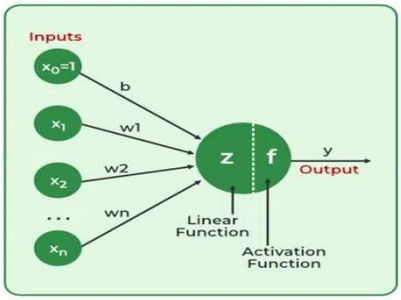
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Formal Definition

- At its core, a neuron is a mathematical function that maps inputs to an output.

$$y = f\left(\sum_{i=1}^{n} w_i x_i + b\right)$$



- Where:
  - $\boldsymbol{x}$ **(Inputs)**: The input vector (raw data or signals from previous layers).
  - $\boldsymbol{w}$ **(Weights):** The learnable parameters that determine the strength of the connection.
  - $\boldsymbol{\Sigma}$**(Summation):** The linear combination of inputs and weights.
  - $\boldsymbol{b}$ **(Bias):** An adjustable threshold that shifts the activation function.
  - $\boldsymbol{f}$ **(Activation Function):** Decide if the neuron should "fire", it can be logistic (0|1), linear or non-linear functions

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The Inputs and Weights

Inputs

- The raw features of your data, represented as a vector $x = [x_1, x_2, \ldots x_n]$.
- Examples
  - Credit Card Fraud: x_1 = Amount spent, x_2= Location, x_3= Time.
  - Image Recognition: x_1= Pixel 1 brightness, x_2= Pixel 2 brightness...

Weights

- Each input x_i has a corresponding weight w_i, the learnable parameters that determine the signal strength.
- Weights are like volume tuner
  - **Positive Weight**: This feature is very important *(e.g., "Has Wings" for identifying a bird).*
  - **Negative Weight**: This feature strongly argues against the class *(e.g., "Has Wheels" for identifying a bird).*
  - **Zero Weight**: This feature is irrelevant *(e.g., "Has eyes" for identifying the bird).*

The model cannot change the inputs; it can change how it reacts to them through tuning weights

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The Weighted Sum and Bias

Sum

- The core operation of the neuron is the Dot Product.

$$z = (x_1 \cdot w_1) + (x_2 \cdot w_2) + \cdots + (x_n \cdot w_n)$$

- This z represents the total "evidence" collected by the neuron.
  - High z: Strong evidence to activate.
  - Low z: Weak evidence.

Bias

- An extra parameter added to the sum, independent of the inputs.
- Why do we need it?
  - If all inputs are 0 the sum is always 0, no matter the weights, so without a bias, the neuron becomes "stuck".

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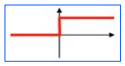
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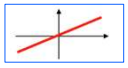
The Activation Function

- Activation function: A mathematical equation that determines the output of a neural network model.
  - Sit at the end of the perceptron, receiving the weighted sum (z) and transforming it into the final output (y).
  - Decide whether a neuron should be activated

Neuron output
- Logistic: transform Z to 1/Yes or 0/No
- Linear: transform with linear f(z) function to create linear output

- The original perceptron used a simple threshold logic to activate the neuron (called step function)
  - If $z > 0$: Output 1 (Neuron is activated).
  - otherwise: Output 0 (Neuron stays inactive).


Logistic: Step (0 or 1)


Linear: f(z)

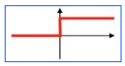
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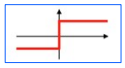
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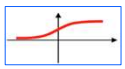
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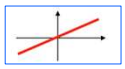
The Activation Function in Examples

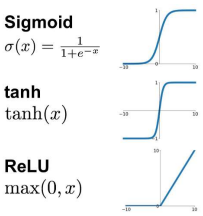
- Basic activation functions
  - **Step**: gets triggered above a certain value of z; else it outputs zero.
  - **Sign/Unit**: outputs +1 or -1 depending on whether z > 0 or not.
  - **Sigmoid**: the S-curve and outputs a value between 0 and 1 (linear)
- Other functions, still just a small subset


Step (0 or 1)


Sign (-1 or 1)


Sigmoid (from 0 to 1)


Linear (f(z))

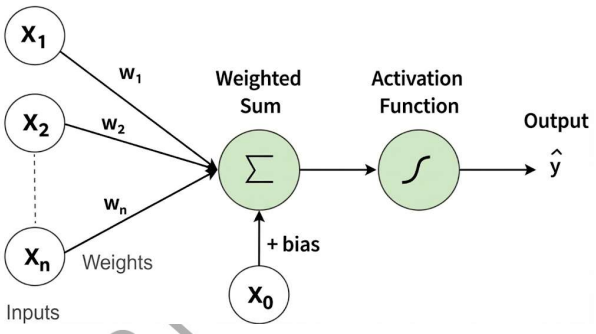


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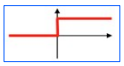
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Perceptron in Review



$$y = \begin{cases} 1 & \text{if } w \cdot x + b > 0 \\ 0 & \text{otherwise} \end{cases}$$



Step (0 or 1)

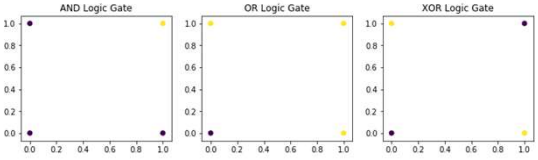
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Logic Gates with Perceptron

Logic gates: AND OR XOR

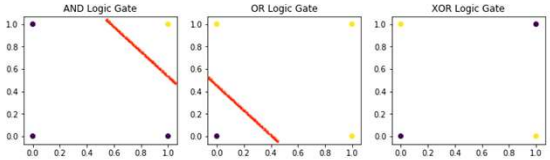
Name	AND	OR	XOR
Symbol			
Truth Table	A B X 0 0 0 1 0 0 0 1 0 1 1 1	A B X 0 0 0 1 0 1 0 1 1 1 1 1	A B X 0 0 0 1 0 1 0 1 1 1 1 0



Logistic regression problem: can we find a linear line f(x) to separate the data points of AND, OR, XOR?

XOR Problem

- AND/OR is a **linear problem**, can solve by a single perceptron



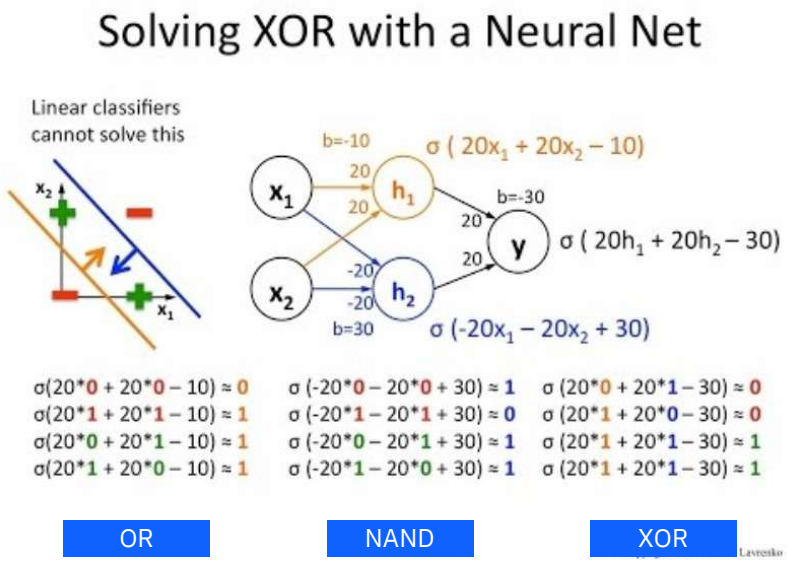
- XOR is **non-linear problem**, can not be solved by a single perceptron, but multiple perceptrons (MLP) can

ML Project: Perceptron Logic Gates - Adrian Pardo

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Lab 7.1: XOR Problem Explained



https://www.youtube.com/watch?v=kNPGXgzxoHw

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- Single Perceptron Model
- Multi-Layer Perceptrons (MLP)

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Definition about Deep Learning

> **Deep Learning** allows computational models that are composed of multiple processing layers to learn representations of data with multiple levels of abstraction.
>
> *LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. Nature*

- The "**Deep**" refers to the depth of the network (**hidden layers**).
  - **Shallow Network**: 1 or 2 hidden layers.
  - **Deep Network**: Dozens or even hundreds of hidden layers.
- It is an attempt to mimic the human brain's architecture to learn from large amounts of data.

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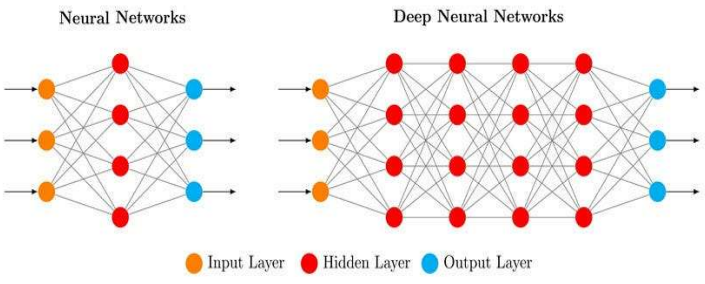
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Hidden Layers for deep models

- A single-layer perceptron can only learn simple, linear relationships between inputs and outputs

Hidden layers extend perceptrons from linear decision makers to powerful models capable of learning complex, non-linear relationships.

- Hidden layers allow the model to
  - Solve complex problems that a single perceptron cannot
  - Learn non-linear patterns by combining multiple neurons and activation functions
  - Extract higher-level features from raw inputs
- Each hidden layer transforms the input into a more useful representation, making it easier for the final layer to make accurate predictions.

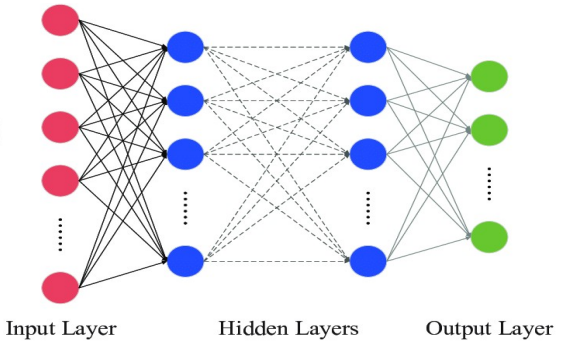


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MLP for Deep neuron networks (DNN)

- Components:
  - **Input Layer:** The raw facts.
  - **Hidden Layers:** Neurons here process features in each layer
  - **Output Layer:** The final decision.
- Terms:
  - MLP is also called as Deep Neural Network (DNN) or Deep feed-forward networks (FFNN)
  - MLP is one type of Deep Learning model, but Deep Learning includes many other architectures beyond MLP.



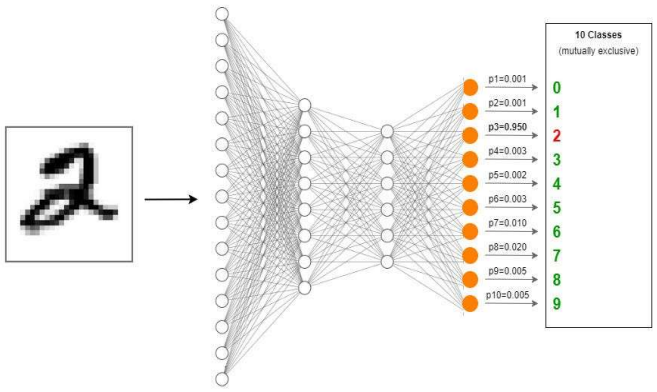
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Case Study: Detecting hand-written number

- Input layer (the flattening)
  - We take the 28x28 grid image and "unroll" it into a single long list of 784 input neurons.
- Hidden layers:
  - The network learns patterns based on pixel values, but it is rigid.
  - The diagram uses 2 hidden layers
- Output layer:
  - 10 neurons representing the probability for each digit (0–9).

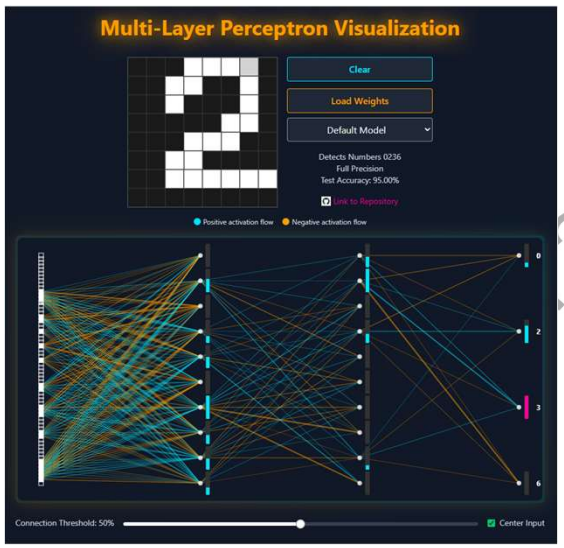


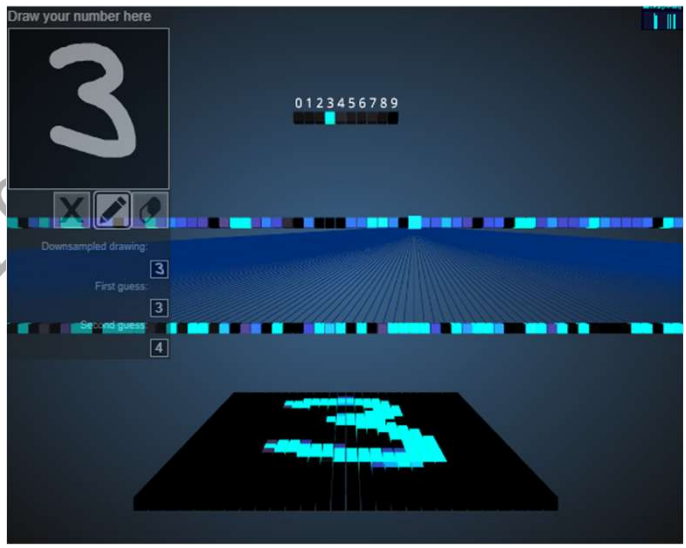
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MLP Simulation



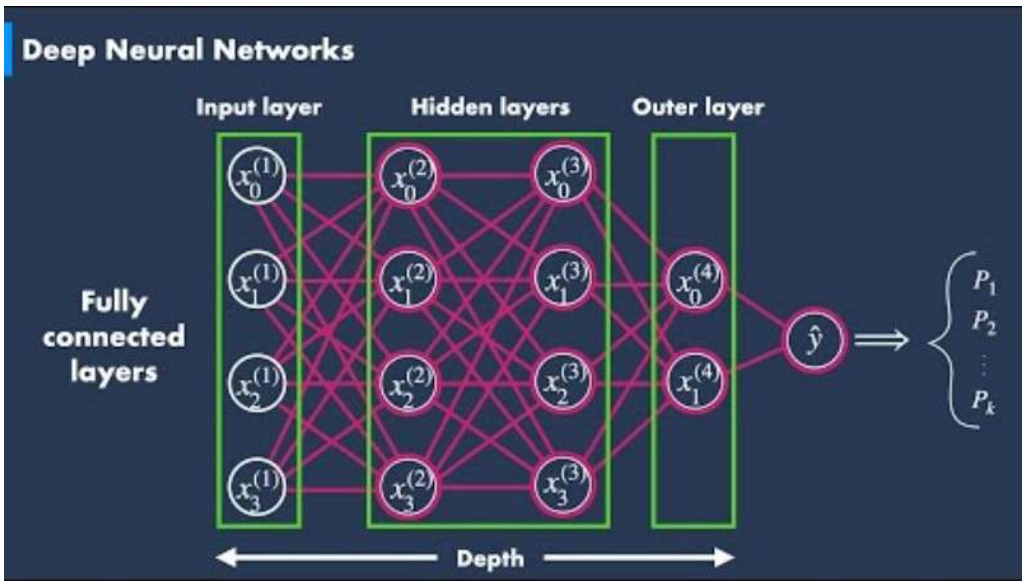


https://cpldcpu.github.io/neural-network-visualizer/ https://adamharley.com/nn_vis/mlp/3d.html

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MLP Video



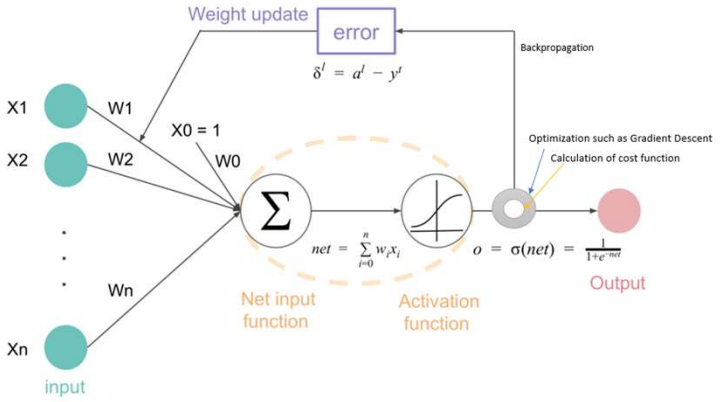
https://www.youtube.com/watch?v=pbQtQ2Bdzf8 (9m)

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Training a MLP/DNN

- Step 1: Forward Propagation
  - Data flows from Input →Hidden → Output.
  - The network makes a prediction based on current (initially random) weights.
- Step 2: Loss Calculation
  - Compare the Prediction ($\hat{y}$) vs. Actual Label (y).
  - Calculate the error (Loss) using a function like Mean Squared Error or Cross-Entropy.
- Step 3: Backpropagation
  - Calculate the gradient: "Which weight contributed most to this error?"
- Step 4: Optimization
  - Adjust the weights slightly in the opposite direction of the error (Gradient Descent).
  - Repeat thousands of times.



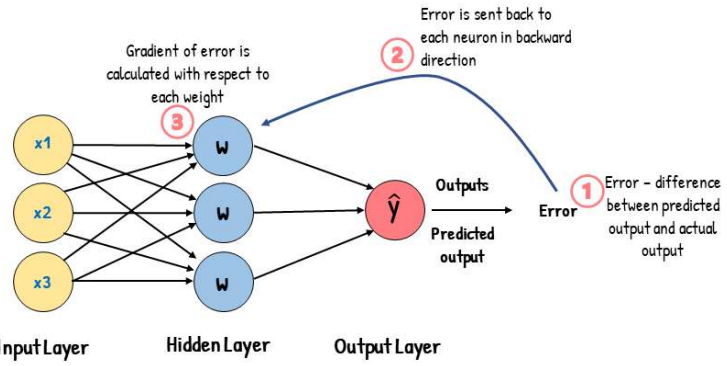
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Loss optimization with Backpropagation

Backpropagation is formally defined as an algorithm for supervised learning of ANN that calculates the gradient of a loss function with respect to the network's weights

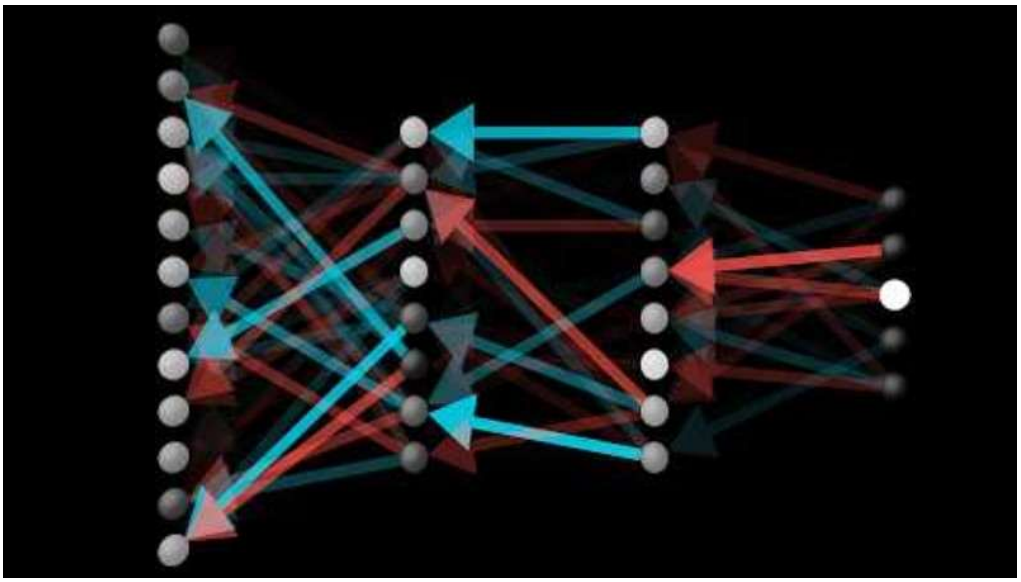


Backpropagation: short for backward propagation of errors

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Backpropagation in review



https://www.youtube.com/watch?v=Ilg3gGewQ5U&t=460s (13m)

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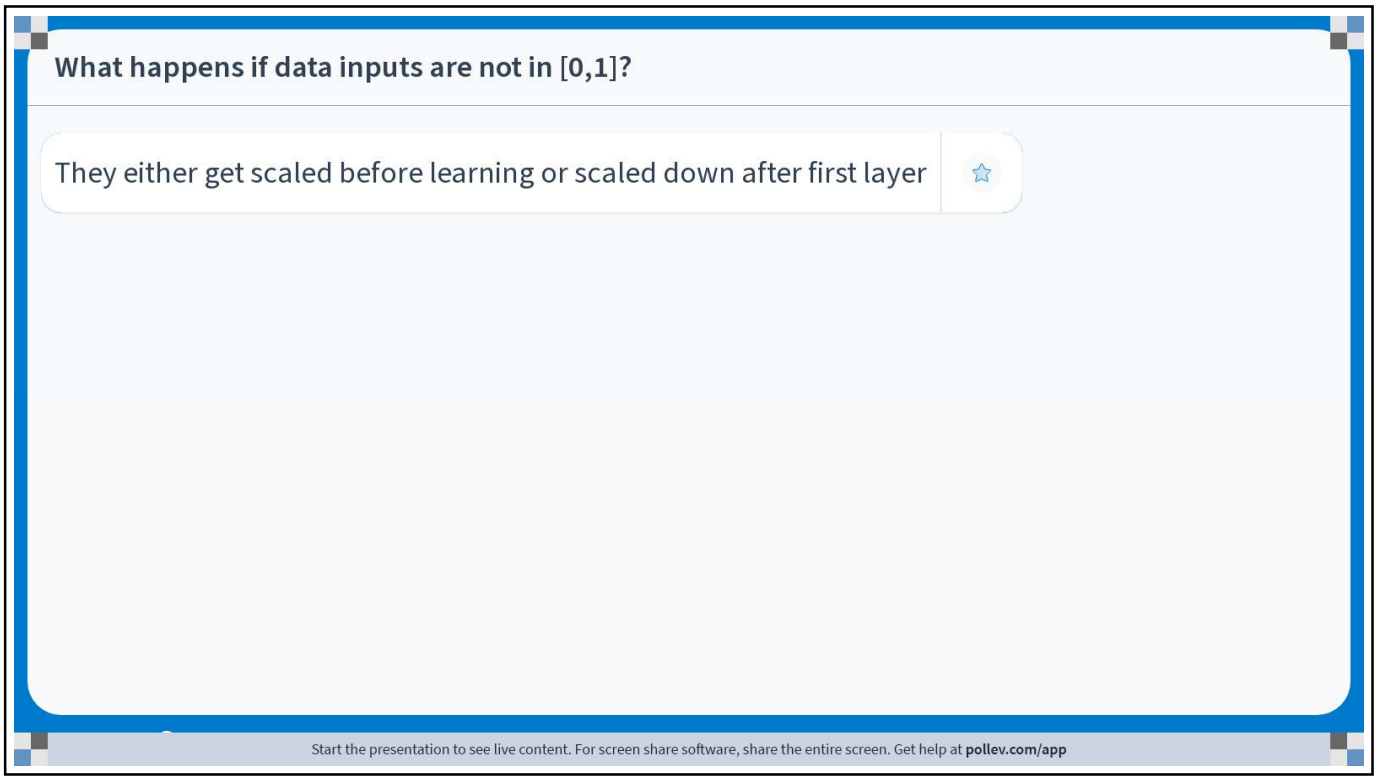
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Why Perceptron (Single, MLP and Basic Deep Learning) prefers working number between 0 and 1?

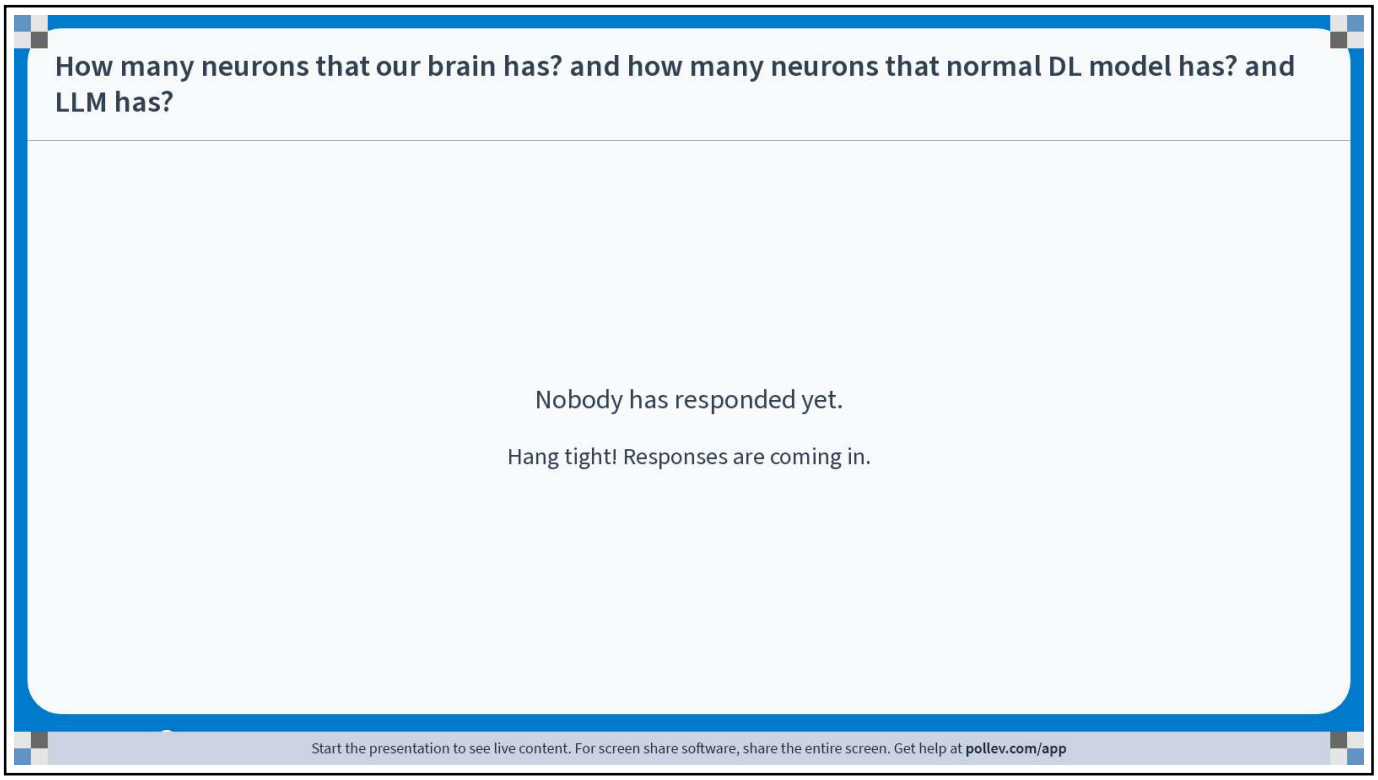
Because final output is percentage

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What happens if data inputs are not in [0,1]?
They either get scaled before learning or scaled down after first layer
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How many neurons that our brain has? and how many neurons that normal DL model has? and LLM has?
Nobody has responded yet.
Hang tight! Responses are coming in.
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Some numbers about neurons

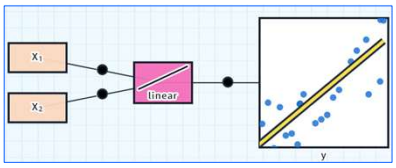
Brain/AI	Neurons
Human	~85-90 billions
Mouse	~71 millions
Fly	~100k
Simple DNN/FFNN	~1-10k
Large CNN	~1m
Standard GPT	~1-5m

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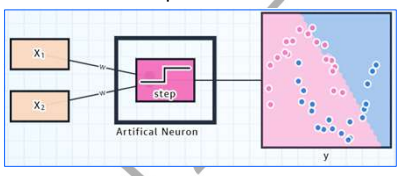
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Recap: From ML to NN

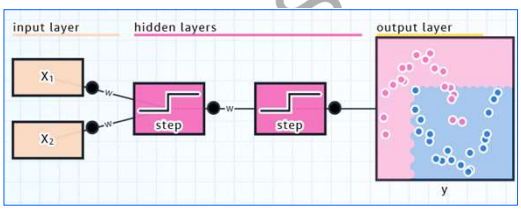
Linear



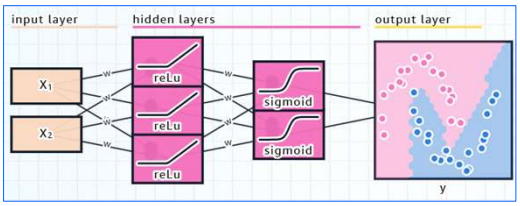
Perceptron



MLP



FFNN



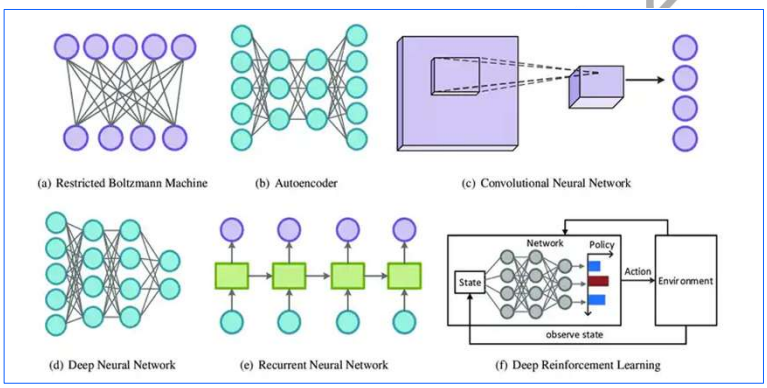
https://mlu-explain.github.io/neural-networks/

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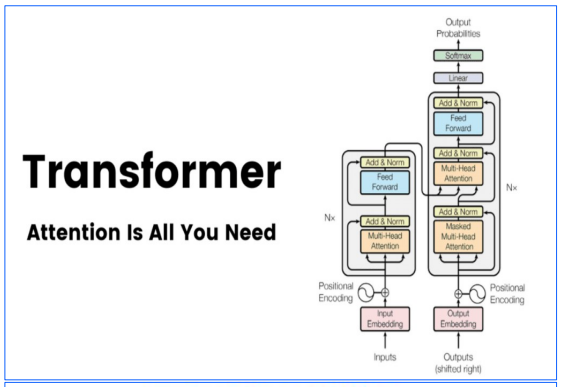
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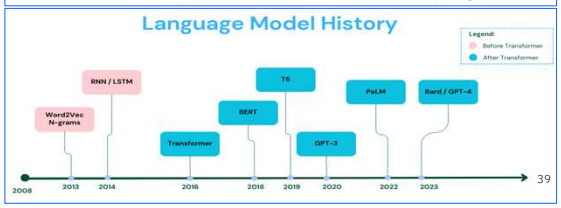
Deep Learning Families for next Lessons

- From MLP/DNN to other Deep Learning models
  - An Deep MLP with many hidden layers (DNN). The deeper the network, the more complex things it can understand.
  - The Deep MLP is good at tabular data, our models need enhancement to learn more complex data like images, vision, time series, language...



- Transformers (2016) - The Age of LLM





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Thank you!

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